

Customer Churn Risk Analysis

Executive Summary

Project Overview

This analysis examines customer churn patterns within a telecommunications company to identify high-risk customer segments and opportunities for improving retention.

Using data analysis techniques including feature engineering, SQL aggregation, and interactive visualization, the project identifies behavioral, contractual, and pricing factors that influence churn probability.

The objective of this analysis is to help leadership prioritize customer retention efforts and reduce revenue loss.

Data & Methodology

The analysis was conducted using a telecom customer dataset containing customer demographics, service usage, billing information, and churn status.

The workflow included:

- Data cleaning and feature engineering in spreadsheets
- Risk score model creation based on customer attributes
- SQL queries to analyze churn drivers
- Dashboard visualization using Tableau Public

A churn risk scoring model was created to segment customers based on contract type, tenure, billing level, and service usage indicators.

Key Findings

1. Contract Type is the Strongest Churn Predictor

Customers on month-to-month contracts churn at significantly higher rates than customers on long-term contracts.

2. Early Lifecycle Customers Are High Risk

Customers within their first 12 months show the highest churn probability, suggesting onboarding experience plays a major role in retention.

3. High Monthly Charges Increase Churn Risk

Customers paying above-average monthly charges demonstrate increased churn rates, indicating potential pricing sensitivity.

4. Risk Score Model Demonstrates Predictive Value

Customers with higher risk scores show a consistent increase in churn probability, validating the usefulness of the segmentation model.

Strategic Recommendations

1. Incentivize Contract Upgrades

Encourage month-to-month customers to transition to annual contracts through targeted discounts or loyalty incentives.

2. Improve New Customer Onboarding

Introduce proactive engagement for customers within their first 12 months to reduce early lifecycle churn.

3. Monitor High-Risk Customers

Use the churn risk score to flag customers likely to churn and prioritize them for retention outreach.

4. Evaluate Pricing Strategy

Analyze pricing sensitivity among high-charge customers to determine whether alternative pricing plans could improve retention.

Potential Business Impact

By identifying high-risk customer segments and implementing targeted retention strategies, organizations can reduce churn and preserve recurring revenue.

Even a modest reduction in churn rates can generate significant long-term revenue improvements for subscription-based businesses.

Interactive Dashboard

An interactive dashboard visualizing the analysis and key metrics can be accessed here:

[Tableau Dashboard Link](#)